

Author	Study area	Study period	Water quality parameter	Sensor	Method	Bands	Application	Classification
Caballero et al. (2022)	Mar menor lagoon (Spain)	2021	Turbidity	Landsat 8	Empirical Models	Red	Technical and environmental: spatiotemporal distribution of turbidity and chlorophyll-a concentration	Technical with some environmental application.
Bonansea et al. (2015)	Río Tercero reservoir (Argentina)	2003-2010	Temperature	Landsat 5	Empirical Models	Thermal	Technical and environmental: quantitative maps providing spatially and temporally information on patterns of water quality throughout the reservoir. Water quality features related to the hydrogeomorphology of the reservoir, typical seasonality and influx from the cooling system of a local nuclear reactor were identified in the time series maps.	Technical with some environmental application.
Bonansea et al. (2015)	Río Tercero reservoir (Argentina)	2003-2010	Chl-a	Landsat 5	Empirical Models	Blue and green	Technical and environmental: quantitative maps providing spatially and temporally information on patterns of water quality throughout the reservoir. Water quality features related to the hydrogeomorphology of the reservoir, typical seasonality and influx from the cooling system of a local nuclear reactor were identified in the time series maps.	Technical with some environmental application.
Bonansea et al. (2015)	Río Tercero reservoir (Argentina)	2003-2010	Transparency	Landsat 5	Empirical Models	NIR and Blue	Technical and environmental: quantitative maps providing spatially and temporally information on patterns of water quality throughout the reservoir. Water quality features related to the hydrogeomorphology of the reservoir, typical seasonality and influx from the cooling system of a local nuclear reactor were identified in the time series maps.	Technical with some environmental application.
Chao Rodríguez et al. (2013)	Lake Arreo (Spain)	2000-2012	Temperature	Landsat 5	Empirical Models	Thermal	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Chao Rodríguez et al. (2013)	Lake Arreo (Spain)	2000-2012	Chl-a	Landsat 5	Empirical Models	Blue and green	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Chao Rodríguez et al. (2013)	Lake Arreo (Spain)	2000-2012	Transparency	Landsat 5	Empirical Models	Blue and green	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Hicks et al. (2013)	34 lakes (New Zealand)	2000-2009	TSS	Landsat 5	Semi-empirical models	NIR	Technical	Purely technical
Hicks et al. (2013)	34 lakes (New Zealand)	2000-2009	Turbidity	Landsat 5	Semi-empirical models	NIR	Technical	Purely technical
Hicks et al. (2013)	34 lakes (New Zealand)	2000-2009	Transparency	Landsat 5	Semi-empirical models	Blue and red	Technical	Purely technical
Kutser (2012)	Lake Malären (Sweden)	1984–2011	CDOM	Landsat 5	Semi-empirical models	green and red	Climate Change	Purely technical
Lobo, Costa, and Novo (2015)	Tapajós River Basin (Brazil)	1973-2013	SPM	Landsat 5	Semi-empirical models	Red	Relation to gold mining activities; spatio-temporal information on water quality	Environmental with trend analysis
Mancino et al. (2009)	Monticchio lakes (Italy)	2001	Chl-a	Landsat 5	Empirical Models	Blue, green and red	Technical	Purely technical
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Simon, Tormos, and Danis (2014)	Lake Bariousses, Lake Bimont (France)	2006-2013	Temperature	Landsat 5	Semi-Analytical Models	Thermal	Suitability of Landsat archive for information on historical lake Tsurf	Technical with some environmental application.
Imen, Chang, and Y. J. Yang (2015)	Lake Mead (United States)	2005-2009	TSS	Landsat 5	Machine Learning Models	Not reported	Construct a real-time forecasting model for predicting degraded water quality near drinking water outlets	Environmental with trend analysis
Lymburner et al. (2016)	19 lakes (Australia)	1988-2013	TSS	Landsat 5	Semi-Analytical Models	Not reported	Examines interactions between decadal climate variations (ENSO) and TSS concentrations in optically heterogenous lakes across western Australia.	Technical with some environmental application.
Concha and Schott (2016)	Long Pond and Cranberry Pond (United States)	3 dates (2013, 2014, 2015)	Chl-a	Landsat 8	Semi-empirical models	Coastal Aerosol	Shows that the Landsat 8 satellite can be utilized over Case 2 waters as long as a careful atmospheric correction is applied.	Technical with some environmental application.
Concha and Schott (2016)	Long Pond and Cranberry Pond (United States)	3 dates (2013, 2014, 2015)	TSS	Landsat 8	Semi-empirical models	Coastal Aerosol	Shows that the Landsat 8 satellite can be utilized over Case 2 waters as long as a careful atmospheric correction is applied.	Technical with some environmental application.
Concha and Schott (2016)	Long Pond and Cranberry Pond (United States)	3 dates (2013, 2014, 2015)	CDOM	Landsat 8	Semi-empirical models	Coastal Aerosol	Shows that the Landsat 8 satellite can be utilized over Case 2 waters as long as a careful atmospheric correction is applied.	Technical with some environmental application.
Shahzad et al. (2018)	Indus Delta Region (Pakistan)	November 6, 2001; March 22, 2005; and March 20, 2016	TSS	Landsat 7	Empirical Models	Blue, green red, NIR and SWIR	Technical	Purely technical
Malahlela (2019)	5 Dams (South Africa)	2014 - 2018	TSS	Landsat 8	Semi-empirical models	Not reported	Technical	Purely technical
C. Wang et al. (2018)	Pearl River Estuary (China)	1987–2015	TSS	Landsat 5-9	Semi-empirical models	Near infrared and red	Analyze the spatial and temporal evolution of TSS concentrations in Pearl River Estuary (PRE).	Environmental with trend analysis
J. Chen et al. (2017)	Lake Huron (Canada-United States)	2013	CDOM	Landsat 5	Semi-empirical models	Red and NIR	Technical	Purely technical

J. Wang and X. Chen (2024)	Erhai Lake (China)	2013 - 2021	Chl-a	Landsat 8	Machine Learning Models	16 Spectral Index, Visible and NIR	Technical: tested 4 atmospheric correction procedures and Spectral index and single bands; Also evaluated general trend, showing that there was an insignificant decreasing trend in the mean annual Chl-a concentration from 2003 to 2021 ($R^2 = 0.10$, $p = 0.16$), indicating a reduction in the eutrophication level and a slightly improvement in water quality. This is likely related to lake protection policies implemented by the local government	Technical with some environmental application.
He et al. (2021)	Dashahe Reservoir (China)	2013-2018 (2 images/year)	Chl-a	Landsat 8	Machine Learning Models	1 to 7	Eutrophication was evaluated: The Dashahe Reservoir was basically in the state of mesotrophication and light eutrophication. The area of light eutrophication accounted for larger proportions in spring and autumn, and the reservoir inflow was the main source of nutrient salts.	Environmental trend analysis with
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He et al. (2021)	Dashahe Reservoir (China)	2013-2018 (2 images/year)	Phosphorous	Landsat 8	Machine Learning Models	1 to 7	Eutrophication was evaluated: The Dashahe Reservoir was basically in the state of mesotrophication and light eutrophication. The area of light eutrophication accounted for larger proportions in spring and autumn, and the reservoir inflow was the main source of nutrient salts.	Environmental trend analysis with
Dave et al. (2019)	Narmada River and Reserviors (Bargi, Ukai, and Ujjani) (India)	2017-2018	Chl-a	Landsat 8	Semi-empirical models	Blue, green and red	Technical	Purely technical
Dave et al. (2019)	Narmada River and Reserviors (Bargi, Ukai, and Ujjani) (India)	2017-2018	Turbidity	Landsat 8	Semi-empirical models	Blue, green and red	Technical	Purely technical
Bangira et al. (2024)	Olushandja Dam (Namibia)	2014- 2015	Turbidity	Landsat 8	Semi-empirical models	Red, blue and green and NIR	There is substantial spatial variation in the concentration of water quality parameters both longitudinally and latitudinally due to several factors: Rainfall, nutrient accumulation, human activities such as fishing, abstraction of water and swimming	Environmental trend analysis with
Bangira et al. (2024)	Olushandja Dam (Namibia)	2014- 2015	TSS	Landsat 8	Semi-empirical models	Red, blue and green and NIR	There is substantial spatial variation in the concentration of water quality parameters both longitudinally and latitudinally due to several factors: Rainfall, nutrient accumulation, human activities such as fishing, abstraction of water and swimming	Environmental trend analysis with
Bangira et al. (2024)	Olushandja Dam (Namibia)	2014- 2015	Phosphorous	Landsat 8	Semi-empirical models	Red, blue and green and NIR	There is substantial spatial variation in the concentration of water quality parameters both longitudinally and latitudinally due to several factors: Rainfall, nutrient accumulation, human activities such as fishing, abstraction of water and swimming	Environmental trend analysis with
Masocha et al. (2018)	Two reservoirs (Zimbabwe)	June 2013	Transparency	Landsat 8	Semi-empirical models	Blue and red	Water quality status for Lake Chivero has significantly deteriorated, when compared to Lake Mazvikadei. Lake Chivero is highly eutrophic and this can be attributed to the lake's proximity to Harare Metropolitan City: Poor waste disposal mechanism	Environmental trend analysis with
Masocha et al. (2018)	Two reservoirs (Zimbabwe)	June 2013	Turbidity	Landsat 8	Semi-empirical models	Blue and red	Water quality status for Lake Chivero has significantly deteriorated, when compared to Lake Mazvikadei. Lake Chivero is highly eutrophic and this can be attributed to the lake's proximity to Harare Metropolitan City: Poor waste disposal mechanism	Environmental trend analysis with
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Mohsen, Elshemy, and Zeidan (2021)	Lake Burullus (Egypt)	2010-2013	Chl-a	Landsat 7	Semi-empirical models	Blue and red	Water quality of Lake Burullus is in a critical state, especially the southern part of the lake due to the drains effluents and the reclamation activities	Technical with some environmental application.
Mohsen, Elshemy, and Zeidan (2021)	Lake Burullus (Egypt)	2010-2013	TSS	Landsat 7	Semi-empirical models	Blue and SWIR	Water quality of Lake Burullus is in a critical state, especially the southern part of the lake due to the drains effluents and the reclamation activities	Technical with some environmental application.
Mohsen, Elshemy, and Zeidan (2021)	Lake Burullus (Egypt)	2010-2013	Temperature	Landsat 7	Semi-empirical models	Thermal	water quality of Lake Burullus is in a critical state, especially the southern part of the lake due to the drains effluents and the reclamation activities	Technical with some environmental application.
Guo, X. Zhu, et al. (2023)	Lake Simcoe (Canada)	2013-2019	Chl-a	Landsat 8	Machine Learning Models	Bands 1-5, OC2, OC3, BGR, NDCI, CVI	Calculated the correlations between the WQPs and the climate factors (i.e., air temperature, pressure, precipitation, wind speed, and wind direction), the anthropogenic factors (i.e., urban and farmland areas, nighttime light, and the number of mobile devices connecting to cellular towers (hereinafter termed Cell for brevity)), and other potential factors (i.e., the normalized difference vegetation index (NDVI) and discharge)	Environmental with trend analysis
Guo, X. Zhu, et al. (2023)	Lake Simcoe (Canada)	2013-2019	Phosphorous	Landsat 8	Machine Learning Models	Bands 1-5, OC2, OC3, BGR, NDCI, CVI	Calculated the correlations between the WQPs and the climate factors (i.e., air temperature, pressure, precipitation, wind speed, and wind direction), the anthropogenic factors (i.e., urban and farmland areas, nighttime light, and the number of mobile devices connecting to cellular towers (hereinafter termed Cell for brevity)), and other potential factors (i.e., the normalized difference vegetation index (NDVI) and discharge)	Environmental with trend analysis
Guo, X. Zhu, et al. (2023)	Lake Simcoe (Canada)	2013-2019	Transparency	Landsat 8	Machine Learning Models	Bands 1-5, OC2, OC3, BGR, NDCI, CVI	Calculated the correlations between the WQPs and the climate factors (i.e., air temperature, pressure, precipitation, wind speed, and wind direction), the anthropogenic factors (i.e., urban and farmland areas, nighttime light, and the number of mobile devices connecting to cellular towers (hereinafter termed Cell for brevity)), and other potential factors (i.e., the normalized difference vegetation index (NDVI) and discharge)	Environmental with trend analysis
Bernardo, do Carmo, et al. (2019)	Six reservoirs (Brazil)	2014 y 2016	TSS	Landsat 8	Semi-Analytical Models	Green and red	Technical	Purely technical
Du et al. (2020)	Songnen Plain (China)	1984 to 2018	TSS	Landsat 5, 7 and 8	Semi-empirical models	Red/Blue	In Songnen Plain, the mean annual TSM in 60.5% of the water bodies decreased from 1984 to 2018. The decreasing of TSM is likely due to the increasing of vegetation in the area. The TSM concentration in waters declined from April to July, and then increased from September onwards. We also found the TSM in water bodies in Songnen Plain has very high spatial variation. Our results indicated that the meteorological factors such as wind and precipitation may affect the variation of TSM.	Environmental with trend analysis
H. Yang et al. (2022)	Baiyangdian lake (China)	July 2019 - September 2019	Chl-a	Landsat 8	Machine Learning Models	Not reported	Technical: inversion model of Chl-a concentration based on CNN with fused high spatial and temporal resolution images	Technical with some environmental application.
Arias-Rodriguez, Z. Duan, et al. (2021)	Five lakes (Mexico)	2012-2018	Chl-a	Landsat 8	Machine Learning Models	Not reported	Technical: implications of the lack of integration of remote sensing in water quality monitoring routines	Purely technical
Arias-Rodriguez, Z. Duan, et al. (2021)	Five lakes (Mexico)	2012-2018	Turbidity	Landsat 8	Machine Learning Models	Not reported	Technical: implications of the lack of integration of remote sensing in water quality monitoring routines	Purely technical
Arias-Rodriguez, Z. Duan, et al. (2021)	Five lakes (Mexico)	2012-2018	TSS	Landsat 8	Machine Learning Models	Not reported	Technical: implications of the lack of integration of remote sensing in water quality monitoring routines	Purely technical
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Arias-Rodriguez, Tüzün, et al. (2023)	Global (United States, France, Italy, Germany, Greece, Norway, Poland, England, Ireland, Mexico, Canada, Japan, Chile, Russia, Cyprus, Sweden, Finland)	2021-2022	Chl-a	Landsat 8	Machine Learning Models	FERT: Original bands + red/green + NIR, (green + red)/2, green/red, red/green, NIR/green + latitude, longitude, year and month	Technical: developed machine learning models for water quality retrieval at a global scale using remote homogenized multimodal remote sensing data. This contributed to overcoming the present state of knowledge in which the transferability of models is limited by the origin of field data, and modeling water quality in inland waters at different locations was constrained	Technical with some environmental application.
Arias-Rodriguez, Tüzün, et al. (2023)	Global (United States, France, Italy, Germany, Greece, Norway, Poland, England, Ireland, Mexico, Canada, Japan, Chile, Russia, Cyprus, Sweden, Finland)	2021-2022	Turbidity	Landsat 8	Machine Learning Models	FERT: Original bands + red/green + NIR, (green + red)/2, green/red, red/green, NIR/green + latitude, longitude, year and month	Technical: developed machine learning models for water quality retrieval at a global scale using remote homogenized multimodal remote sensing data. This contributed to overcoming the present state of knowledge in which the transferability of models is limited by the origin of field data, and modeling water quality in inland waters at different locations was constrained	Technical with some environmental application.
Arias-Rodriguez, Tüzün, et al. (2023)	Global (United States, France, Italy, Germany, Greece, Norway, Poland, England, Ireland, Mexico, Canada, Japan, Chile, Russia, Cyprus, Sweden, Finland)	2021-2022	TSS	Landsat 8	Machine Learning Models	FERT: Original bands + red/green + NIR, (green + red)/2, green/red, red/green, NIR/green + latitude, longitude, year and month	Technical: developed machine learning models for water quality retrieval at a global scale using remote homogenized multimodal remote sensing data. This contributed to overcoming the present state of knowledge in which the transferability of models is limited by the origin of field data, and modeling water quality in inland waters at different locations was constrained	Technical with some environmental application.
Arias-Rodriguez, Tüzün, et al. (2023)	Global (United States, France, Italy, Germany, Greece, Norway, Poland, England, Ireland, Mexico, Canada, Japan, Chile, Russia, Cyprus, Sweden, Finland)	2021-2022	Transparency	Landsat 8	Machine Learning Models	FERT: Original bands + red/green + NIR, (green + red)/2, green/red, red/green, NIR/green + latitude, longitude, year and month	Technical: developed machine learning models for water quality retrieval at a global scale using remote homogenized multimodal remote sensing data. This contributed to overcoming the present state of knowledge in which the transferability of models is limited by the origin of field data, and modeling water quality in inland waters at different locations was constrained	Technical with some environmental application.
Arias-Rodriguez, Tüzün, et al. (2023)	Global (United States, France, Italy, Germany, Greece, Norway, Poland, England, Ireland, Mexico, Canada, Japan, Chile, Russia, Cyprus, Sweden, Finland)	2021-2022	Phosphorous	Landsat 8	Machine Learning Models	FERT: Original bands + red/green + NIR, (green + red)/2, green/red, red/green, NIR/green + latitude, longitude, year and month	Technical: developed machine learning models for water quality retrieval at a global scale using remote homogenized multimodal remote sensing data. This contributed to overcoming the present state of knowledge in which the transferability of models is limited by the origin of field data, and modeling water quality in inland waters at different locations was constrained	Technical with some environmental application.
H. Zhang et al. (2022)	Dongping Lake (China)	2013-2018	Chl-a	Landsat 8	Machine Learning Models	all bands and different ratios	Used statistical analysis to identify the spatial and temporal heterogeneity. one-way analysis of variance (ANOVA) was applied in this study to investigate the spatiotemporal patterns	Technical with some environmental application.
H. Zhang et al. (2022)	Dongping Lake (China)	2013-2018	Phosphorous	Landsat 8	Machine Learning Models	all bands and different ratios	Used statistical analysis to identify the spatial and temporal heterogeneity. one-way analysis of variance (ANOVA) was applied in this study to investigate the spatiotemporal patterns	Technical with some environmental application.
Markogianni, Kalvas, et al. (2018)	Trichonis Lake (Greece)	October 2013 and August 2014	Chl-a	Landsat 8	Semi-empirical models	Coastal aerosol, blue and green	Technical: use of remote sensing technology and specifically of Landsat 8 OLI sensor, to accurately quantify certain water quality parameters.	Purely technical
Qiang Wang, Kaishan Song, Wen, et al. (2021)	Hulun Lake (China)	1986 - 2017	TSS	Landsat 7 and 8	Semi-empirical models	NIR	The mean TSM value of Hulun Lake shows an increasing trend in the innerlake area and a decreasing trend in the outer lake from 1986 to 2017. relationship of TSM to water level, water area, precipitation, and wind speed were analysed	Environmental with trend analysis
W. Zhu et al. (2020)	West Lake (China)	2013 - 2017	TSS	Landsat 8	Semi-empirical models	Red and NIR	TSM presented diverse distributions in different sections and months/seasons—the relatively high-concentration TSM was usually found in the north section and during spring festivals, holidays and summer vacation. These spatiotemporal patterns demonstrate the evidence that TSM in West Lake was highly linked to anthropogenic impacts, such as tourist visits, commercial activities, and urban drainage projects	Technical with some environmental application.

Abdelal et al. (2022)	King Talal Dam (Jordan)	2018 - 2019	Chl-a	Landsat 8	Semi-empirical models	Red and NIR	Technical: Compared different atmospheric correction methods. Shows high effect of the adjacent land on the obtained Chl-a concentration relative to the much deeper water near the dam	Technical with some environmental application.
Abdelal et al. (2022)	King Talal Dam (Jordan)	2018 - 2019	Turbidity	Landsat 8	Semi-empirical models	Coastal aerosol and green	Technical: Compared different atmospheric correction methods.	Purely technical
Lehmann et al. (2019)	23 lakes in Rotorua Lakes Region (New Zealand)	1999 - 2018	Transparency	Landsat 7 and Landsat 8	Empirical Models	Blue, red and green	More environmental than technical: A trend analysis on annual and summertime averages of Secchi depth revealed several statistically significant and positive trends, indicating improving water clarity	Environmental with trend analysis
D. Tian et al. (2024)	Qianwu reservoir, Longjing reservoir and Nanshan reservoir (China)	2021-2022	Chl-a	Landsat 7	Semi-empirical models	Blue, red and green	Proposed an inversion strategy of water quality variables in various monitoring frequencies and weather conditions were proposed finally.	Environmental with trend analysis
D. Tian et al. (2024)	Qianwu reservoir, Longjing reservoir and Nanshan reservoir (China)	2021-2022	TSS	Landsat 7	Semi-empirical models	Red, NIR and SWIR	Proposed an inversion strategy of water quality variables in various monitoring frequencies and weather conditions were proposed finally.	Environmental with trend analysis
D. Tian et al. (2024)	Qianwu reservoir, Longjing reservoir and Nanshan reservoir (China)	2021-2022	Turbidity	Landsat 7	Semi-empirical models	Green, red and NIR	Proposed an inversion strategy of water quality variables in various monitoring frequencies and weather conditions were proposed finally.	Environmental with trend analysis
Y. Yang et al. (2023)	Lake Dianchi (China)	1998 - 2020	Chl-a	Landsat 5 and 8	Machine Learning Models	Blue, green, red and NIR	Most technical: include texture index. The result shows that the Caohai has a higher degree of eutrophication than the Waihai, and the water quality in the subregions of Lake Dianchi showed an improvement trend, suggesting the robustness of the model.	Purely technical
Zeng et al. (2023)	4 Yunnan Plateau lakes (China)	January 2013–August 2022	Transparency	Landsat 8	Empirical Models	Not reported	Technical	Purely technical
Zeng et al. (2023)	4 Yunnan Plateau lakes (China)	January 2013–August 2022	Transparency	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Dallosch and Creed (2021)	Lakes (United States and Sweden)	1984–2016	Chl-a	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Technical and environmental: to classify optical water types using a guided unsupervised classification approach in which OWTs are defined through both remotely sensed reflectance and surface water chemistry data taken from samples in North American and Swedish lakes	Technical with some environmental application.
Robert et al. (2017)	Agoufou Lake (Mali)	2000-2017	TSS	Landsat 7 and 8	Semi-empirical models	NIR and red	The satellite estimates are then employed to assess the SPM dynamics in the main lakes and ponds of the Gourma region and its links to environmental and anthropogenic factors.	Technical with some environmental application.
El-Alem and Chokmani (2022)	Lakes in Quebec (Canada)	2014-2017	Turbidity	Landsat 7 and 8	Machine Learning Models	Red, Blue and SWIR1	Technical	Purely technical
Erena et al. (2019)	Mar menor lagoon (Spain)	2015-2017	Turbidity	Landsat 8	Semi-empirical models	Green and red	This provided the information needed to perform a time series analysis of the lagoon. We also analyzed the processes that occur in the salt lagoon, characterizing the different spatiotemporal patterns of biophysical parameters. Special attention was given to the role of turbidity and chlorophyll-a levels in the Mar Menor ecosystem with regard to the programs of integral management of this natural space that receives maximum environmental protection. T	Environmental with trend analysis
Erena et al. (2019)	Mar menor lagoon (Spain)	2015-2017	Chl-a	Landsat 8	Semi-empirical models	Green and red	This provided the information needed to perform a time series analysis of the lagoon. We also analyzed the processes that occur in the salt lagoon, characterizing the different spatiotemporal patterns of biophysical parameters. Special attention was given to the role of turbidity and chlorophyll-a levels in the Mar Menor ecosystem with regard to the programs of integral management of this natural space that receives maximum environmental protection. T	Environmental with trend analysis
Oliphant et al. (2018)	The Vaal Dam (South Africa)	2014–2016	Chl-a	Landsat 8	Semi-empirical models	Green, red and NIR	The resultant maps indicate that higher class concentrations were predominantly found at or near the shores of the dam. These areas are usually shallower with higher temperatures than the deep waters, resulting in lateral diurnal thermal variations	Technical with some environmental application.

E. S. Deutsch, Fortin, and Cardille (2022)	Lakes across southern (Canada)	2017–2020	Transparency	Landsat 8	Empirical Models	Blue and red	Combined with ancillary information on lake morphological, hydrological, and watershed geological and landuse characteristics, we were able to assess broad spatial patterns in water clarity for the first time. Ecological zones, underlying geological substrate, and lake depth had particularly strong influences on clarity across the whole country.	Technical with some environmental application.
E. S. Deutsch, Cardille, et al. (2021)	Lakes across southern (Canada)	2013–2019	Transparency	Landsat 8	Semi-empirical models	Blue and red	Technical	Purely technical
Yunlin Zhang et al. (2016)	Xin'anjiang Reservoir (China)	December 2013 - April 2015	TSS	Landsat 8	Semi-Analytical Models	Red	The spatial heterogeneity, significantly higher in incoming rivers than the main body of the reservoir, was due to watershed inputs and anthropogenic dredging activity. The temporal heterogeneity of TSM, significantly higher in summer and autumn than in winter and spring, was mainly caused by the seasonal rainfall and seasonal growth of phytoplankton	Technical with some environmental application.
Yibo Zhang et al. (2021)	Various lakes (China)	2016 - 2018	Transparency	Landsat 5, 7 and 8	Semi-empirical models	Red	Technical	Purely technical
Markogianni, Kalivas, et al. (2020)	50 lake water bodies (Greece)	2013–2016 and 2018	Chl-a	Landsat 7 and 8	Semi-empirical models	Blue, red and SWIR	Principal component analysis (PCA) is performed to explore Greek lakes' potential interrelationships based on their Chl-a values in conjunction with certain criteria: their characteristics (artificial/natural), typology, and climatic type	Technical with some environmental application.
Villota-González et al. (2023)	Cajititlán and Zapotlán lakes (Mexico)	2013–2023	Chl-a	Landsat 8	Machine Learning Models	Green, red and NIR	Technical	Purely technical
Villota-González et al. (2023)	Cajititlán and Zapotlán lakes (Mexico)	2013–2023	TSS	Landsat 8	Machine Learning Models	Green, red and NIR	Technical	Purely technical
Villota-González et al. (2023)	Cajititlán and Zapotlán lakes (Mexico)	2013–2023	Turbidity	Landsat 8	Machine Learning Models	Green, red and NIR	Technical	Purely technical
Coelho et al. (2017)	Marengo, Paus Branco and São Nicolau reservoirs (Brazil)	2014–2015	Chl-a	Landsat 8	Semi-empirical models	green and NIR	Tecnical and Limnological Dynamics and Trophic Conditions	Technical with some environmental application.
Coelho et al. (2017)	Marengo, Paus Branco and São Nicolau reservoirs (Brazil)	2014–2015	CDOM	Landsat 8	Semi-empirical models	Green	Tecnical and Limnological Dynamics and Trophic Conditions	Technical with some environmental application.
Meng, J. Zhang, and Zheng (2022)	Shanmei Reservoir (China)	2020 - 2022	Turbidity	Landsat 8–9	Semi-empirical models	Blue, green and red	Explored the relationship between different WQPs using correlation analysis and principal component analysis (PCA)	Environmental with trend analysis
Meng, J. Zhang, and Zheng (2022)	Shanmei Reservoir (China)	2020 - 2022	Chl-a	Landsat 8–9	Semi-empirical models	Blue, green and red	Explored the relationship between different WQPs using correlation analysis and principal component analysis (PCA)	Environmental with trend analysis
Yin et al. (2022)	Lake Taihu (China)	1984 - 2020	TSS	Landsat 5 and 8	Semi-empirical models	Red, green and NIR	Downward trend over the study period, which might be primarily attributable to a decline in wind speed	Technical with some environmental application.
Guo, S. Tian, et al. (2022)	Lake Simcoe (Canada)	1984 - 2020	Chl-a	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Reconstructed the historical spatiotemporal patterns of Chl-a, TP, and TN in Lake Simcoe since 1984, and investigated the effects of two water quality improvement programs	Environmental with trend analysis
Guo, S. Tian, et al. (2022)	Lake Simcoe (Canada)	1984 - 2020	Phosphorous	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Reconstructed the historical spatiotemporal patterns of Chl-a, TP, and TN in Lake Simcoe since 1984, and investigated the effects of two water quality improvement programs	Environmental with trend analysis
Cui et al. (2022)	Chaohu Lake, Shengjin Lake, and Jiaogang Lake (China)	2013–2019	Transparency	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Chegoonian et al. (2023)	Buffalo Pound Lake (Canada)	2014–2020	Chl-a	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Bernardo, Carmo, et al. (2020)	Tietê River Cascade System (Brazil)	2014 and 2016	TSS	Landsat 8	Semi-empirical models	Blue and green	Technical	Purely technical
Z. Cao et al. (2022)	Yunnan–Guizhou Plateau (China)	2013 to 2022	CDOM	Landsat 8 and 9	Machine Learning Models	Green	Technical with general seasonal analysis: the studied area exhibited apparent seasonal variability from 2013 to 2022, particularly after 2017	Technical with some environmental application.
Alcântara, Curtarelli, and Stech (2016)	Barra Bonita hydroelectric reservoir (Brazil)	May and October 2014	CDOM	Landsat 8	Semi-empirical models	Blue and red	Technical	Purely technical
M. Chen et al. (2022)	Honghu Lake (China)	1987–2020	Transparency	Landsat 5, 7 and 8	Semi-Analytical Models	Red	Technical with general trend analysis: ZSD showed an overall decreasing trend from 1987–2020	Technical with some environmental application.
C. Zhao et al. (2022)	Hulun Lake (China)	2010 to 2020	Transparency	Landsat 5, 7 and 8	Semi-empirical models	Red, Blue and NIR	technical: seasonal, annual and spacial trend	Technical with some environmental application.
Markogianni, Dimitriou, and Karaouzas (2014)	Koumoundourou Lake (Greece)	2003 - 2011	Chl-a	Landsat 5	Semi-empirical models	Green and NIR	Characterization of trophic state	Technical with some environmental application.
J. Li et al. (2018)	Lake Huron (Canada–United States)	2013 and 2015	CDOM	Landsat8	Semi-Analytical Models	Green	Environmental: general seasonal phenology, anual and spacial trends	Environmental with trend analysis
Kutser et al. (2016)	Nohipalu Mustjärv, Meelva, and Mustjärv lakes (Estonia)	July 2013	Chl-a	Landsat8	Semi-empirical models	NIR	Studied Chl-a in extreme (black) lakes where the water reflectance values are negligible in almost entire visible part of spectrum (400–700 nm) due to the absorption by CDOM	Technical with some environmental application.
L. Wang et al. (2020)	12 lakes (United States)	2013 - 2017	Chl-a	Landsat 7 and 8	Machine Learning Models	Not reported	Technical	Purely technical

Watanabe et al. (2015)	Barra Bonita reservoir (Brazil)	2013-2014	Chl-a	Landsat8	Semi-empirical models	Green and NIR	Technical and estimated Chl-a concentration was used to classify the trophic level from a trophic state index	Technical with some environmental application.
Peterson, Sagan, and Sloan (2020)	Several waterbodies in Midwestern (United States)	2013–2018	Chl-a	Landsat8	Machine Learning Models	all bands and different ratios	Technical and use the Empirical Data Analytics (EDA) anomaly detection method on the time series to identify abnormal data points	Technical with some environmental application.
Peterson, Sagan, and Sloan (2020)	Several waterbodies in Midwestern (United States)	2013–2018	Turbidity	Landsat8	Machine Learning Models	all bands and different ratios	Technical and use the Empirical Data Analytics (EDA) anomaly detection method on the time series to identify abnormal data points	Technical with some environmental application.
Ledesma et al. (2019)	Cassaffouth reservoir (Argentina)	2016 -2017	Chl-a	Landsat8	Semi-empirical models	2, 4, 6, 7 and 8	Technical and basic spatial distribution maps to interpretate of water quality conditions in the whole study reservoir	Technical with some environmental application.
Amado-Alvarez et al. (2019)	Virgenes dam, the Colina lake, and La Boquilla dam (Mexico)	22 de marzo de 2011	Turbidity	Landsat 5	Semi-empirical models	Green and red	Technical and basic spatial distribution maps to interpretate of water quality conditions	Technical with some environmental application.
Hafeez et al. (2022)	Eight lakes (China)	2013-2019	TSS	Landsat8	Semi-empirical models	Not reported	Technical	Purely technical
F. Zhang et al. (2021)	Guanting Reservoir (China)	1985 -2019	Chl-a	Landsat 5, 7 and 8	Semi-empirical models	NIR and Blue	Technical and monthly and inter-annual changes, also relationship with temperature, precipitation, wind, and water area	Technical with some environmental application.
Koll-Egyed, Cardille, and E. Deutsch (2021)	128 Southern lakes (Canada)	2017 - 2019	CDOM	Landsat 8	Semi-empirical models	Blue, green and red	Technical	Purely technical
Lu et al. (2023)	107 lakes and reservoirs in Yangtze Plain (China)	1984 - 2020	Transparency	Landsat 5, 7 and 8	Semi-empirical models	Blue, green and red	Technical and general trend	Technical with some environmental application.
K. Song, G. Liu, et al. (2020)	2759 lakes (China)	2013–2018	Transparency	Landsat 8	Semi-empirical models	Blue and red	Technical and national scaled, geographic distribution	Technical with some environmental application.
Z. Yang et al. (2023)	Taihu Lake	2018 - 2022	Turbidity	Landsat 8 and 9	Machine Learning Models	NIR	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Nascimento Silva et al. (2017)	Reservoir of Tocantins river (Brazil)	2007 - 2014	Chl-a	Landsat 7	Machine Learning Models	Not reported	Technical	Purely technical
Nascimento Silva et al. (2017)	Reservoir of Tocantins river (Brazil)	2007 - 2014	Transparency	Landsat 7	Machine Learning Models	Not reported	Technical	Purely technical
Nascimento Silva et al. (2017)	Reservoir of Tocantins river (Brazil)	2007 - 2014	TSS	Landsat 7	Machine Learning Models	Not reported	Technical	Purely technical
Rodrigues et al. (2017)	Nav reservoir (Brazil)	2014 -2016	Transparency	Landsat 8	Semi-Analytical Models	Not reported	Technical	Purely technical
W. Li et al. (2022)	Lake Qinghai (China)	1986 to 2020	TSS	Landsat 5, 7 and 8	Empirical Models	Red	Technical and correlation was found between TSM concentration and daily wind speed and hydrological changes	Technical with some environmental application.
Ha et al. (2017)	10 hypertrophic lakes in Hanoi (Vietnam)	2015 - 2016	Chl-a	Landsat 8	Empirical Models	Blue and green	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Papoutsas, Akylas, and D. Hadjimitsis (2014)	Asprokremmos Reservoir (Cyprus)	May 2010–October 2010	Turbidity	Landsat 7	Empirical Models	Red	Technical: Defining the Landsat TM/ETM+ spectral regions in which turbidity can be retrieved in inland waterbodies	Technical with some environmental application.
Jakovljevic, Álvarez-Taboada, and Govedarica (2023)	Main water bodies (Serbia)	1984-2021	TSS	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Technical	Purely technical
Jakovljevic, Álvarez-Taboada, and Govedarica (2023)	Main water bodies (Serbia)	1984-2021	Chl-a	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Technical	Purely technical
Jakovljevic, Álvarez-Taboada, and Govedarica (2023)	Main water bodies (Serbia)	1984-2021	Phosphorous	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Technical	Purely technical
X. Zhao et al. (2021)	Taihu Lake (China)	2017	Chl-a	Landsat 8	Machine Learning Models	Green and red	Technical	Purely technical
Alcántara, Curtarelli, and Steh (2016)	Itumbiara hydroelectric reservoir (Brazil)	2014	TSS	Landsat 8	Semi-Analytical Models	Green	Technical	Purely technical
Ren et al. (2018)	Three Gorges and Dongting Lake (China)	2013 - 2017	Transparency	Landsat 8	Semi-empirical models	Red and green	More environmental: time series of SD and their probable linkage to the Three Gorges Dam	Purely environmental
Adusei et al. (2021)	Owabi reservoir (Ghana)	Dec 2019	Turbidity	Landsat 8	Machine Learning Models	Green, blue and red	Technical	Purely technical
Adusei et al. (2021)	Owabi reservoir (Ghana)	Dec 2019	TSS	Landsat 8	Machine Learning Models	Green, blue , red and SWIR2	Technical	Purely technical
Maciel et al. (2023)	300+ water bodies (global)	1985-2022	Transparency	Landsat 5, 7 and 8	Machine Learning Models	all bands and 2 semi-analytical models	Technical	Purely technical
Lamaro et al. (2013)	Río Tercero (Argentina)	2000-2006	Temperature	Landsat 7	Semi-Analytical Models	Thermal	Technical	Purely technical
Ghirardi et al. (2023)	Gravel and sand Pit lakes in Po River basin (Italy)	1990 to 2021	TSS	Landsat 5, 7 and 8	Empirical Models	Red	More environmental: This study aims to contribute to the analysis of PLs at a basin scale as an opportunity for environmental rehabilitation and river floodplain management	Technical with some environmental application.
Papoutsas, Akylas, and D. Hadjimitsis (2014)	four different Case-2 waters (Cyprus and Greece)	2010	Transparency	Landsat 5	Semi-empirical models	Green/red	Technical	Purely technical
Barreneche et al. (2023)	Continental water bodies (Uruguay)	2004–2020	Chl-a	Transparency	Semi-empirical models	Not reported	Technical	Purely technical
Rubin et al. (2021)	Several lakes (United States)	1984–2017	Transparency	Landsat 5, 7 and 8	Machine Learning Models	NIR, Green, Blue and red		Purely technical
Vinh et al. (2019)	Thac Ba Reservoir (Vietnam)	2018	Chl-a	Landsat 5, 7 and 8	Empirical Models	green and red	Technical and general trend	Technical with some environmental application.
Martins et al. (2018)	Funil hydroelectric reservoir (Brazil)	1995 to 2010	CDOM	Landsat 5	Semi-empirical models	NIR/blue	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Sun et al. (2024)	Pearl River Estuary (China)	1987-2022	TSS	Landsat 5, 7 and 8	Semi-Analytical Models	Not reported	Technical and general trend	Technical with some environmental application.
Martins et al. (2018)	Sobradinho reservoir (Brazil)	2013–2017	Turbidity	Landsat 8	Empirical Models	Red	Environmental: analysis in multi-year drought	Environmental with trend analysis
Olmanson et al. (2020)	10.000+ Minnesota lakes (United States)	2015 and 2016	CDOM	Landsat 8	Semi-empirical models	Green, red and NIR	Environmental: trend analysis and rainfall correlation	Environmental with trend analysis

Surisetty et al. (2018)	Chilika Lagoon (India)	2014	Turbidity	Landsat 8	Empirical Models	Green and red	Technical and general trend	Technical with some environmental application.
Qiang Wang, Kaishan Song, Xiao, et al. (2022)	Several lakes (United States, Canada and Mexico)	1990-2019	Transparency	Landsat 5, 7 and 8	Semi-empirical models	blue/green + blue/red	Technical. Spatial and temporal general distribution	Technical with some environmental application.
N. Cao et al. (2023)	Ebinur Lake (China)	2014 - 2021	CDOM	Landsat 8	Semi-Analytical Models	Coastal aerosol, Blue, Green, Red and NIR	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Tao et al. (2022)	10,814 lakes (China)	1984 - 2018	Transparency	Landsat 5, 7 and 8	Empirical Models	Red and blue	Environmental: spatial pattern and historical trend in lakes' trophic status	Purely environmental
X. Yang et al. (2020)	Donghu Lake (China)	1987-2018	Chl-a	Landsat 5, 7 and 8	Semi-empirical models	green and red	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Somasundaram et al. (2021)	550 reservoirs (Sri Lanka)	2013 to 2020	Transparency	Landsat 8	Semi-Analytical Models	Coastal Aerosol, Blue, Green and Red	Environmental: temporal and spatial distribution and relationship with natural (precipitation, wind and temperature) and anthropogenic (human activity and population density) drivers	Environmental with trend analysis
Markogianni, Kalivas, et al. (2022)	50 lakes (Greece)	2013-2016 and 2018	Chl-a	Landsat 7 and 8	Semi-empirical models	Blue, red, swir2	Technical and environmental: to analyze trophic state trends and season	Technical with some environmental application.
Markogianni, Kalivas, et al. (2022)	50 lakes (Greece)	2013-2016 and 2018	Transparency	Landsat 7 and 8	Semi-empirical models	Blue, red, green and swir2	Technical and environmental: to analyze trophic state trends and season	Technical with some environmental application.
Markogianni, Kalivas, et al. (2022)	50 lakes (Greece)	2013-2016 and 2018	Phosphorous	Landsat 7 and 8	Semi-empirical models	red and swir1	Technical and environmental: to analyze trophic state trends and season	Technical with some environmental application.
Pahlevan et al. (2022)	Global	Not reported	Chl-a	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Pahlevan et al. (2022)	Global	Not reported	TSS	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Pahlevan et al. (2022)	Global	Not reported	CDOM	Landsat 8	Machine Learning Models	Not reported	Technical	Purely technical
Mushtaq, Nee Lala, and Mantoo (2022)	Wular Lake (India)	October 2018	Chl-a	Landsat 8	Semi-empirical models	NIR, Blue and Green	Technical and environmental: to analyze trophic state	Technical with some environmental application.
Mushtaq, Nee Lala, and Mantoo (2022)	Wular Lake (India)	October 2018	Transparency	Landsat 8	Semi-empirical models	NIR	Technical and environmental: to analyze trophic state	Technical with some environmental application.
Tebbs, Remedios, and Harper (2013)	Lake Bogoria (Kenya)	2003-2005	Chl-a	Landsat 7	Empirical Models	NIR	Technical and Chl-a timeseries and maps	Technical with some environmental application.
Wermuth et al. (2022)	Pampulha reservoir (Brazil)	2013-2020	Temperature	Landsat 8	Semi-Analytical Models	Thermal	Technical and characterize temporal and spatial thermal behavior of the reservoir	Technical with some environmental application.
K. Song, Q. Wang, et al. (2022)	Global	1984-2020	Transparency	Landsat 5, 7 and 8	Semi-empirical models	Red and blue	Environmental: temporal and spatial distribution and relationship with natural (precipitation, wind and temperature) and anthropogenic (human activity and population density) drivers	Environmental with trend analysis
F. Yang et al. (2023)	Honghu Lake (China)	2000-2021	Transparency	Landsat 5, 7 and 8	Semi-empirical models	Not reported	Technical and environmental: to analyze trophic state trends and season	Technical with some environmental application.
Setiawan et al. (2019)	Lake Maninjau (Indonesia)	1987-2018	Transparency	Landsat 7	Semi-empirical models	Blue, green and red	Technical and Long-Term SD Changes	Technical with some environmental application.
Yépez et al. (2024)	Laguna Grande de San Pedro de la Paz (Chile)	2014-2022	Chl-a	Landsat 8	Semi-empirical models	green and NIR	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Yépez et al. (2024)	Laguna Grande de San Pedro de la Paz (Chile)	2014-2022	Turbidity	Landsat 8	Semi-empirical models	Red and NIR	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Tanner, Cardall, and Williams (2022)	Utah Lake (United States)	1984 to 2021	Chl-a	Landsat 5, 7 and 8	Semi-empirical models	SWIR1 and Blue	Environmental: focus on non-parametric Mann-Kendall test to evaluate trends.	Environmental with trend analysis
Huang et al. (2015)	Alaska lakes (United States)	July to August 2013	Temperature	Landsat 8	Empirical Models	Thermal	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Tavares et al. (2019)	Lake Manguera (Brazil)	Not reported	Temperature	Landsat 7	Empirical Models	Thermal	Technical.	Purely technical
Prats et al. (2018)	442 inland water bodies (France)	1999-2016	Temperature	Landsat 5 and 7	Semi-Analytical Models	Thermal	Technical.	Purely technical
Kong et al. (2024)	Poyang Lake (China)	2020 -2021	Temperature	Landsat 8	Semi-Analytical Models	Thermal	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Kramer et al. (2023)	Itaipu Reservoir (Brazil)	2015-2017	Temperature	Landsat 8	Semi-Analytical Models	Thermal	Technical.	Purely technical
Huang et al. (2015)	Arctic Lakes (United States)	2013	Temperature	Landsat 8	Semi-Analytical Models	Thermal	Technical.	Purely technical
Bui (2024)	shrimp farms (Vietnam)	1984 - 2024	Temperature	Landsat 5, 7, 8 and 9	Semi-Analytical Models	Thermal	Technical.	Purely technical
S. Liu et al. (2024)	145 small waterbodies (< 1km ²) (Australia)	1994 -2023	CDOM	Landsat 5, 7 and 8	Semi-empirical models	Blue, red, green	Technical and environmental: trends to analyze external factors	Environmental with trend analysis
H. Li et al. (2025)	Lake Balaton (Hungria)	1984 - 2023	Chl-a	Landsat 5, 7 and 8	Machine Learning Models	Not reported	Technical and environmental: trends	Environmental with trend analysis
Aroko, Deus, and Ngereja (2025)	Lake Victoria (Tanzanian)	June 2020	Turbidity	Landsat 8	Machine Learning Models	Not reported	Technical.	Purely technical
Aroko, Deus, and Ngereja (2025)	Lake Victoria (Tanzanian)	June 2020	TSS	Landsat 8	Machine Learning Models	Not reported	Technical.	Purely technical
Ahmad et al. (2025)	Chilika Lagoon (India)	August 2017 – June 2020	Chl-a	Landsat 7 and 8	Machine Learning Models	Blue, red, green, NIR, SWIR1	Technical.	Purely technical
Sánchez Valdivia et al. (2024)	7 Patagonian Lakes (Argentina)	2016 – 2020	CDOM	Landsat 8	Empirical Models	Blue, red, green, NIR	Technical and environmental: trends to analyze external factors	Environmental with trend analysis
Leggesse et al. (2024)	Lake Tana, Ethiopia	2013-2021	Chl-a	Landsat 8	Machine Learning Models	all	Technical. Spatial and temporal general distribution	Technical with some environmental application.

Byrne et al. (2024)	7 lakes across Ethiopia, Kenya, and bordering Kenya, Uganda, and Tanzania	2005 - 2023	Chl-a	Landsat 5, 7, 8 and 9	Semi-empirical models	Blue, red, green	Technical.	Purely technical
Byrne et al. (2024)	7 lakes across Ethiopia, Kenya, and bordering Kenya, Uganda, and Tanzania	2005 - 2023	TSS	Landsat 5, 7, 8 and 9	Semi-empirical models	Blue, red, green, NIR	Technical.	Purely technical
Byrne et al. (2024)	7 lakes across Ethiopia, Kenya, and bordering Kenya, Uganda, and Tanzania	2005 - 2023	Chl-a	Landsat 5, 7, 8 and 9	Semi-empirical models	Blue, red, green	Technical.	Purely technical
Shamloo and Sima (2024)	Lake Urmia (Iran)	2016 - 2017	Chl-a	Landsat 8	Machine Learning Models	Not acceded	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Shamloo and Sima (2024)	Lake Urmia (Iran)	2016 - 2017	Phosphorous	Landsat 8	Machine Learning Models	Not acceded	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Shamloo and Sima (2024)	Lake Urmia (Iran)	2016 - 2017	Transparency	Landsat 8	Machine Learning Models	Not acceded	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Mamun, Hasan, and An (2024)	194 artificial reservoirs (South Korea)	2019	Chl-a	Landsat 8	Machine Learning Models	Not reported	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Mamun, Hasan, and An (2024)	194 artificial reservoirs (South Korea)	2019	TSS	Landsat 8	Machine Learning Models	Not reported	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Mamun, Hasan, and An (2024)	194 artificial reservoirs (South Korea)	2019	Transparency	Landsat 8	Machine Learning Models	Not reported	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Mancino et al. (2009)	Monticchio lakes (Italy)	37012	Chl-a	Landsat 5-9	Empirical Models	Blue, red, green	Technical.	Purely technical
Mancino et al. (2009)	Monticchio lakes (Italy)	37012	Transparency	Landsat 5-9	Empirical Models	Blue, red, green	Technical.	Purely technical
Scarpace, Fisher, and Holmquist (1979)	20 winsonsin lakes (united states)	1972 - 1973	Turbidity	Landsat 4	Empirical Models	NIR	Technical.	Purely technical
Doña et al. (2014)	Albufera y Parque regional del Sureste (Spain)	1985-2006	Chl-a	Landsat 5 y 7	Empirical Models	green and NIR	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Doña et al. (2014)	Albufera y Parque regional del Sureste (Spain)	1985-2006	Transparency	Landsat 5 y 7	Empirical Models	green	Technical. Spatial and temporal general distribution	Technical with some environmental application.
Domínguez Gómez, Chuvieco Salinero, and Sastre Merlín (2009)	Parque Regional Sureste (Spain)	1984 -2000	Transparency	Landsat 7	Empirical Models	green	Technical. Spatial and temporal general distribution	Technical with some environmental application.
D. J. Carpenter and S. M. Carpenter (1983)	fresh lakes (Australia)	1978 - 1979	Turbidity	Landsat 4	Empirical Models	Not reported	Technical.	Purely technical
Büttner et al. (1987)	Lake Balaton and reservoir Kisköre (Hungria)	Not reported	Chl-a	Landsat 5	Empirical Models	Not reported	Technical.	Purely technical
Büttner et al. (1987)	Lake Balaton and reservoir Kisköre (Hungria)	Not reported	TSS	Landsat 5	Empirical Models	Not reported	Technical.	Purely technical
Dekker, Vos, and Peters (2001)	Frisian lake system (Netherland)	1995	TSS	landsat 5	Empirical Models	red and green	Technical.	Purely technical
Zhou et al. (2006)	Lake Taihu (China)	Not reported	TSS	landsat 5	Empirical Models	Red and NIR	Technical.	Purely technical
D. G. Hadjimitsis et al. (2006)	Kourris Dam (Cyprus)	1998 -2001	Turbidity	landsat 5 and 7	Empirical Models	Red	Technical.	Purely technical
Sudheer, Chaubey, and Garg (2006)	Beaver Reservoir (united states)	2001 - 2004	Chl-a	Landsat 5	Machine Learning Models	Blue, red, green and NIR	Technical.	Purely technical
Wu et al. (2008)	Poyang Lake (China)	2004 - 2005	Transparency	Landsat 5	Empirical Models	Not acceded	Technical.	Purely technical
H. Duan, Y. Zhang, et al. (2008)	Three inland lakes (China)	2004 - 2005	Chl-a	Landsat 5	Empirical Models	red and green	Technical.	Purely technical
H. Duan, Ma, et al. (2009)	Three lakes, Chagan, Xinmiao, and Kuli (China)	2004 - 2005	Transparency	Landsat 5	Empirical Models	red and green	Technical.	Purely technical
Onderka, Rodný, and Velísková (2011)	Embalse Gabčíkovo (Slovakia)	36647	TSS	Landsat 5	Semi-Analytical Models	Blue, red, green	Technical.	Purely technical

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